

Deliverable D11 · WP3

Seminar – Report

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Principal investigator: Ing. Roman Budjač, PhD.

Seminar report

The seminar was held as part of Work Package WP3 (Development and Implementation of a Training Strategy) and focused on possible solutions to the upcoming challenge. Attention was deliberately focused on two questions: how to compare the weight distributions of the reference and quantized models, and how to turn the result into a usable signal represented by a single similarity index. The goal of the seminar was not to present a single solution, but to present and compare several options and, based on them, define a recommended direction for implementation. This report summarizes the technical issues that were the subject of the presentation.

1. Subject and aim of the seminar

The D12 implementation rests on a core task: to compare the weight distribution of the full-precision (FP32) reference model with that of its quantised counterpart and to turn this difference into a numerical indicator — the similarity index. The seminar broke this task down into three decisions, for which it presented alternatives — the choice of the statistical comparison measure, the way the index is used, and the granularity of the quantisation itself - and concluded them with a recommended combination. Such an approach makes it possible to justify the implementation design transparently and to leave room for future extensions.

2. Option A

The first option is the choice of the measure with which the weight histograms, built on the same bin grid, are compared. Four candidate measures were presented. The Kullback–Leibler divergence is sensitive and unbounded and reveals information loss well, but it is asymmetric. The Jensen–Shannon divergence is symmetric and bounded to the interval $[0, 1]$, and is therefore practical for thresholding and automated decision-making. Cosine similarity compares the overall orientation of the distributions and is very favourable in terms of

computational cost. The Wasserstein distance approaches the difference geometrically and handles even distributions that do not overlap. JS divergence is recommended as the primary measure precisely because of its boundedness and symmetry. Distributional moments and entropy serve as a complement to characterise the shape of the distribution.

3. Option B

The second decision is the way the index is used in practice; the seminar examined it from four aspects. In terms of granularity, one can compute a global index over the weights of the whole model, or a per-layer index that reveals the most sensitive layer. In terms of its role in training, the index can serve as a predictor before training (for selecting the bit-width), or as a monitor during training (evaluated per epoch). In terms of the reference, the weight distribution can be compared against a frozen FP32 reference, or between consecutive epochs. And finally, in terms of decision-making, a fixed threshold (for example $JS \leq \tau$) can be used to select the bit-width and to stop training early once the index stabilises.

4. Option C

The third decision is the granularity of the quantisation itself, which also affects what the index is able to capture. Per-tensor quantisation uses a single shared scale for the entire kernel of a layer; it is simple and fast, but for deep networks with weight outliers it can lead to a collapse of accuracy. Per-channel quantisation assigns a separate scale to each output channel; at the cost of slightly higher computational demands, it is significantly more robust and near-lossless for deep networks. The choice of granularity is therefore not merely a technical detail but substantially changes the behaviour of the whole pipeline.

5. Recommended for D12

Based on the comparison of the variants, the seminar defined a recommended combination for the D12 implementation:

- use JS divergence as the primary measure (with KL divergence and cosine similarity as complementary), compute the index both globally and per layer, use it in both roles, as a predictor and as a monitor and quantise per-channel for robustness with deep networks;
- other options such as the Wasserstein distance, an epoch-over-epoch reference and activation monitoring remain open options for further research and development.

6. Next Steps

Following the seminar, the chosen direction will be implemented as the D12 code and subsequently validated on various datasets and architectures. Ongoing evaluation will make it possible to incorporate some of the proposed extensions as needed.

7. Conclusion

The seminar fulfilled its aim. Instead of a single solution, it clearly presented and compared the options for comparing distributions and using the index, and on that basis defined a clear and well-justified direction for the implementation. The meeting thus established a coherent basis for the implementation phase of the project.

Ing. Roman Budjač, PhD.